MIL-I-50364(MU)
19 November 1973
SUPERSEDING
RP-PD-120
15 May 1954

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED: M8 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated: M8 Assembly, loaded with Cartridge Impulse, M68 for use in conjunction with other Propellant Actuated Devices, for aricraft emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect. on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military	
MIL-A-2550	- Ammunition and Special Weapons, General Specifications for

STANDARDS

Military	
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS-28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS

Armament Command	
F10534132	- Initiator, Cartridge Actuated, M8 Assembly
C8796726	- Container, Fiber, Ammunition M380 for Initiator, Cartridge Actuated, M8 with Cartridge, M68
C8796707	- Box, Packing, Ammunition for Initiator Cartridge Actuated, M8, with Cartridge, M68 in Fiber Container, M380

FSC 1377

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DRAWINGS (Cont'd)

Armament Command

C9241015	- Carton, Alternative Packing, Ammunition, for Initiator, Cartridge Actuated, M8 with Cartridge, M68 in Fiber Container, M380
IEL10534132	- Inspection Equipment List for Initiator, Cartridge Actuated, M8 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and rereferenced specifications. (See 6.1)

3.2 Firing pin release.-The firing pin of the assembled initiator, less safety pin, cartridge, body, filter and cap, shall release from the cocked position when the application of the axial load of 25 to 35 pounds has withdrawn the initiator pin 3/4 inches, maximum.

3.3 Initiator pin security.-The initiator pin of the assembled initiator less cartridge, body, filter and cap, shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall not cause permanent deformation of the safety pin.

3.4 Breakaway torque.-The initiator cap and body, and the body and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.-The loaded initiator shall not deflect the indicator of a compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.6 Functional requirements.

3.6.1 Ballistics.-Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values and delay times specified in Table I. Pressure values shall be produced at the end of a 30 foot length of aircraft hose, size #4 in accordance with MS-28741.

TABLE I

Temperature Fahrenheit (F)	Pressure (PSI) Min.	Max.	Ignition Delay (Max) (See 6.3)
-65° to -70°	1000	3000	65 Milliseconds
65° to 75°	1000	3000	65 Milliseconds
160° to 165°	1000	3000	65 Milliseconds

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3.6.2 The initiator shall not fail to fire.

3.6.3 The initiator shall function without mechanical failure (see 6.2).

3.7 Workmanship.-The requirements for workmanship are as specified on the applicable drawing, referenced specification and the following:

3.7.1 Metal defects.-The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item on which might be injurious to personnel handling the item.

3.7.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requriements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submittal of product.-The contractor shall submit the product in accordance with MIL-STD-105.

4.2.1.2 Size of lot.-A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

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4.2.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group the criteria in Table III shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III
Total Allowable Defects Per Sample Size

Sample Size for Major Defects - - - - -	110	150	225	300
No. of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects - - - - -	110	150	225	300
No. of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.2.2.1 Cartridge, Impulse, M68, Drawing D8593492, covering a detail of Drawing F10534132.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. O-ring missing	Visual
4. Sealing disc missing or damaged	Visual
5. Marking missing, incorrect or illegible	Visual
6. Any parts missing	Visual
MAJOR	
101. O-ring split, cracked, or cut	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.2 Assembly, Drawing F10534132 less drawings: F10534136 (Housing, Firing Pin), D10523310 (Body), C10522667 (Filter), D10522668 (Cap), B8359536 (Shipping Cap), and D8593492 (Cartridge, Impulse).

CRITICAL	
1. Firing pin missing	Visual
2. Initiator not in cocked position	Visual
3. Safety pin missing or improperly engaged	Visual
4. Firing pin incorrectly assembled	Visual
5. Firing spring missing	Visual
6. Metal defective	Visual
MAJOR	
101. Thread damaged or not full	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.3 Assembly, Drawing F10534132 less drawings: C10522667 (Filter), D10522668 (Cap), and B8359536 (Shipping Cap).

Defect	Method of Inspection
CRITICAL	
1. Cartridge missing	Visual
2. Cartridge loose	Visual
3. Foreign matter in set screw hole (3 holes)	Visual
4. Foreign matter in firing pin hole	Visual
5. Thread damaged or not full	Visual
6. "O" ring missing	Visual
7. Improper assembly	Visual
8. Set screw missing (3 set screws)	Visual
9. Metal defective	Visual
10. No sharp corners on firing pin housing	Visual
MAJOR	
101. Marking missing, incorrect, or illegible	Visual
102. Foreign matter	Visual
103. Sealing compound missing from set screws	Visual
104. Conalite missing from body threads	Visual
105. "O" ring split, cracked, or not lubricated	Visual
106. Protective finish inadequate or defective	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Drawing F10534132.

CRITICAL	
1. Any part missing, except as otherwise classified	Visual/X-Ray *
2. Improper assembly	Visual
3. Foreign matter in gas port	Visual
4. Thread damaged	Visual
5. Set screw missing (2 set screws), loose; above flush or not covered with barrier material	Visual
6. Marking missing, incorrect or illegible	Visual
7. Safety pin deformed or loose	Visual

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4.2.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Total length, max.	Gage
102. Assembly of stand and strap insecure or incorrect	Visual
103. Copolite missing from body thread	Visual
104. Sealant compound missing from set screw (2 set screws)	Visual
105. Protective finish inadequate or defective	Visual
106. Shipping cap missing	Visual
107. Foreign matter	Visual
MINOR	
None defined	

* The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.2.2.5 Fiber Container, Drawing C8796726.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
101. Gap between body and cover, max.	Manual
102. Contents loose	Manual
103. Metal ends loose or distorted	Manual
104. Cut or damaged through all impregnated layers	Visual
105. Initiator improperly packed	Visual
106. Marking missing, incorrect or illegible	Visual
107. Adhesives improperly applied	Visual
108. Tape incomplete or badly wrinkled	Visual
109. Length of tear tab, min.	Visual
110. Color of tape incorrect	Visual

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4.2.2.6 Packing Box, Drawing C8796707.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
101. Contents loose	Manual
102. Hardware missing, loose, broken or improperly engaged	Visual/Manual
103. Contents improperly packed	Visual
104. Box damaged	Visual
105. Contents of box exposed	Visual
106. Strapping missing, loose, broken or improperly assembled	Visual/Manual
107. Marking missing, incorrect or illegible	Visual
108. Number of units incorrect	Visual
109. Car seal missing, unsealed or improperly positioned	Visual

4.2.2.7 Carton, Alternative Packing, Drawing C9241015.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Carton broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete	Visual

4.2.3 Test Methods and Procedures.

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4.2.3.1 Firing pin release.-The initiator pin of each subassembly shall be subjected to the firing pin release test. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in accordance with requirements as specified in 3.2, in conjunction with the fixture which registers the amount of pull. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.-The initiator pin of each subassembly shall be subjected to the initiator pin security test. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in accordance with requirements as specified in 3.3, in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in each position. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.-Each initiator shall be subjected to the breakaway torque application. A torque wrench set at the torque requirement specified in 3.4 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torque shall be applied to the cap. Failure of any initiator assembly to comply with the requirements of 3.4 shall be cause for rejection of the initiator.

4.2.3.4 Residual magnetism.-Each initiator shall be subjected to the test for excessive residual magnetism. The test requires the use of an approved compass, type AN-5766, in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator. Failure of any initiator to comply with the requirements of 3.5 shall be cause for rejection of the initiator.

4.2.3.5 Functional.-The following tests shall be performed by the Government:

4.2.3.5.1 Ballistics.-Twenty-one units shall be tested to determine compliance with the requirements of 3.6.1, 3.6.2, and 3.6.3. The units shall be temperature conditioned as provided in Table IV and 4.2.3.5.1.1. The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.2.3.5.1.1. Failure of any units to comply with the requirements of 3.6.1, 3.6.2, and 3.6.3 shall be cause for rejection of the lot.

TABLE IV

Temperature	Sample Size
-65° to -70°F	7
65° to 75°F	7
160° to 165°F	7

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4.2.3.5.1.1 Temperature conditioning.-A standard M8 Initiator loaded with a dummy M68 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.2.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 8595400.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring activity: Level A - Drawings C8796707 and C8796726, Level B - Dwgs. C8796726 and C9241015.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Applicable initiator drawing number.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurence of which is other than a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

Custodian:

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Preparing Activity:

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-50364(MU) Initiator, Cartridge Actuated: M8 Assembly

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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